

R.V.COLLEGE OF ENGINEERING

(Autonomous Institution Affiliated to VTU)

V Semester B.E

Model Question paper

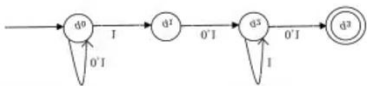
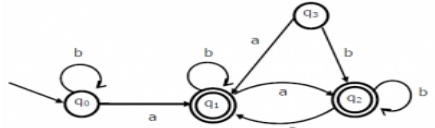
THEORY OF COMPUTATION

Time: 03 Hours

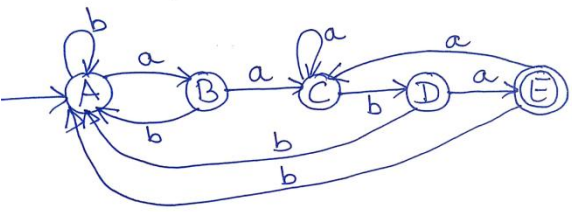
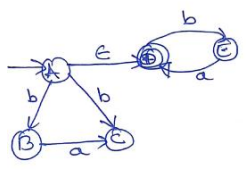
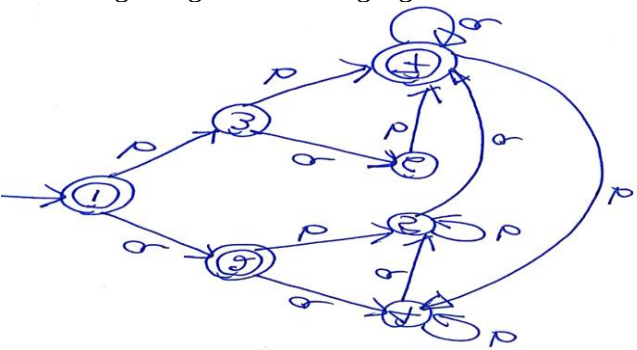
Maximum Marks: 100

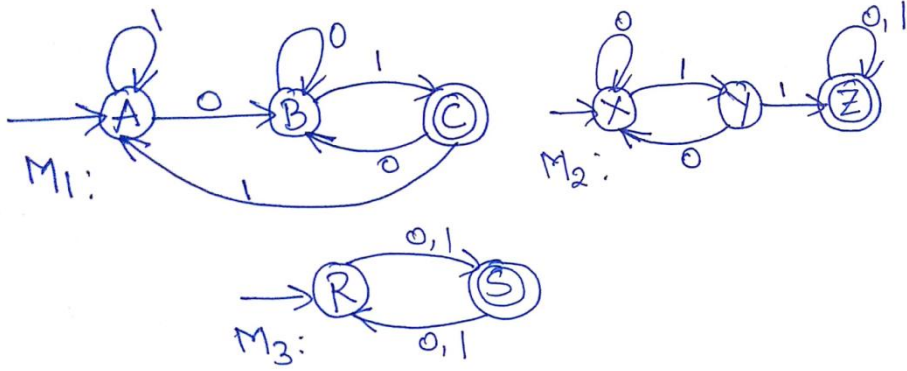
Instructions to Candidate:

1. Answer all questions from PART-A. PART-A questions to be answered in first three pages of the answer book.
2. Answer question 2 and any one question from 3 and 4, any one question from 5 and 6, any one question from 7 and 8 AND any one question from 9 and 10 PART-B.

	PART-A	
1.1	Consider the following automata. What is the set of reachable states for the input string 0011? 	1
1.2	The minimum state automaton equivalent to the below FA has how many number of states 	1
1.3	Find $R^2_{12} = ?$ 	1
1.4	Define ϵ-closure (q), where $q \in Q$ of an automata.	1
1.5	In each case, find a string of minimum length in $\Sigma^* = \{0, 1\}^*$ not in the language corresponding to the given regular expression. i. $(0^* + 1^*)(0^* + 1^*)(0^* + 1^*)$ ii. $0^*(100^*)^*1^*$	1

1.6	Describe the decision algorithm to answer the following question. Given two DFA's M1 and M2, are there any strings that are accepted by neither?	1																		
1.7	Give CFG for the language "The set of odd length strings in $\{a, b\}^*$ with the middle symbol a".	1																		
1.8	Show that the grammar with indicated productions is ambiguous. $S \rightarrow aSb \mid aaSb \mid \epsilon$	1																		
1.9	Define Left Most Derivation (LMD). Give LMD for abaaba in the grammar with productions $S \rightarrow aSaSbS \mid aSbSaS \mid bSaSaS \mid \epsilon$	1																		
1.10	Identify the nullable variables in the grammar given below. $S \rightarrow AB \mid ABC, A \rightarrow BA \mid BC \mid a \mid \epsilon, B \rightarrow AC \mid CB \mid b \mid \epsilon, C \rightarrow AB \mid BC \mid c$	1																		
1.11	If G is a context free grammar in CNF and $x \in L(G)$ with $ x =n$. How many steps are there in the derivation of x in G?	1																		
1.12	Let G be the CFG with productions $S \rightarrow S+S \mid S-S \mid S*S \mid S/S \mid (S) \mid a$ How many derivation strings are there for the string $a+(a*a)/a-a$?	1																		
1.13	DFA whose transition table is as below, here state A is the start state and states B, C are final states. <table border="1" data-bbox="289 835 464 1024"> <tr><td>δ</td><td>a</td><td>b</td></tr> <tr><td>A</td><td>B</td><td>B</td></tr> <tr><td>B</td><td>C</td><td>C</td></tr> <tr><td>C</td><td>D</td><td>D</td></tr> <tr><td>D</td><td>A</td><td>B</td></tr> </table> What is the equivalent left linear grammar for the above DFA.	δ	a	b	A	B	B	B	C	C	C	D	D	D	A	B	1			
δ	a	b																		
A	B	B																		
B	C	C																		
C	D	D																		
D	A	B																		
1.14	Consider the right linear grammar with the productions $A \rightarrow aB \mid bD, B \rightarrow aB \mid bC \mid \epsilon, C \rightarrow aB \mid bC \mid \epsilon, D \rightarrow aD \mid bD$ Find the left linear grammar which is equivalent to the above right linear grammar.	1																		
1.15	Define Deterministic Push Down Automata.	1																		
1.16	What is the language generated by the Context Sensitive Grammar with the productions $S \rightarrow ABCS \mid ABC, AB \rightarrow BA, AC \rightarrow CA, BC \rightarrow CB, BA \rightarrow AB, CA \rightarrow AC, CB \rightarrow BC, A \rightarrow a, B \rightarrow b, C \rightarrow c$.	1																		
1.17	Obtain Turing Machine to accepts the language $L = \{w \mid w \in \{a, b\}^* \text{ and } w \text{ ends with } abb\}$.	1																		
1.18	What is the language generated by the right linear grammar with productions $S \rightarrow aA \mid bC \mid b, B \rightarrow aC \mid bA \mid a, A \rightarrow aS \mid bB, C \rightarrow aB \mid bS$.	1																		
1.19	If L1 and L2 are recursively enumerable languages, then $L1 \cap L2$ is recursively enumerable language. Is this statement is TRUE or FALSE.	1																		
1.20	What is the solution to the instance of PCP given below. <table border="1" data-bbox="289 1617 782 1841"> <tr><td></td><td>List A</td><td>List B</td></tr> <tr><td></td><td>X_i</td><td>Y_i</td></tr> <tr><td>1</td><td>1</td><td>010</td></tr> <tr><td>2</td><td>0</td><td>10</td></tr> <tr><td>3</td><td>10</td><td>101</td></tr> <tr><td>4</td><td>01</td><td>100</td></tr> </table>		List A	List B		X_i	Y_i	1	1	010	2	0	10	3	10	101	4	01	100	1
	List A	List B																		
	X_i	Y_i																		
1	1	010																		
2	0	10																		
3	10	101																		
4	01	100																		

	PART-B	
2 a	Find the regular expressions corresponding to each of the following subsets of $\{0,1\}^*$. a) The language of all strings containing exactly two 0's. b) The language of all strings that do not end with 01.	4
b	For the DFA below describe, either in words or by writing regular expressions, the strings that cause the DFA to be in each state. 	4
c	Explain the algorithm to find an equivalent NFA from the given NFA- ϵ . Use this algorithm to draw an NFA for the NFA- ϵ given below. 	4
d	For the DFA shown below, use the minimization algorithm to find a minimum state DFA recognizing the same language. 	4
3 a	State and prove pumping lemma for regular languages. Use pumping lemma to show that the following languages are not regular. a) $L = \{ww \mid w \in \{0,1\}^*\}$. b) $L = \{xy \mid x, y \in \{0,1\}^* \text{ and } y \text{ is either } x \text{ or } x^r\}$.	6
b	Describe decision algorithm to answer the following questions. a) Given two DFA's M_1 and M_2 , are there any strings that are accepted by neither. b) Given a regular expression r and an DFA M , are the corresponding languages are same?	4

c	Let M1, M2 and M3 are the DFA's pictured below recognizing languages L1, L2 and L3 respectively.  Draw DFA's recognizing the following languages. $L1 \cup L2$ $L1 \cap L2$ $L1 - L2$	6
OR		
4 a	Define CFG, Language generated by CFG, Left most derivation and Right most derivation. Consider the grammar below $S \rightarrow S+S \mid S-S \mid S*S \mid S/S \mid (S) \mid a$. - Give two left most derivations for the string $a+(a*a)/a-a$. - Give the derivation tree corresponds to each of the derivations in (i). - How many distinct left most derivations of this string are there? - How many derivation trees are there for the string $a+a+a+a+a$?	4
b	In each case, given the context free grammar G, find an equivalent CFG with no useless variables. $S \rightarrow ABC \mid BaB, A \rightarrow aA \mid BaC \mid aaa, B \rightarrow bBb \mid a, C \rightarrow CA \mid AC$ $S \rightarrow AB \mid AC, A \rightarrow aAb \mid bAa \mid a \mid \epsilon, B \rightarrow bbA \mid aaB \mid AB, C \rightarrow abCa \mid aDb \mid \epsilon, D \rightarrow bD \mid aC$	4
c	Define GNF grammar. Given below a CFG G, find a CFG G^1 in GNF generating $L(G) - \{ \epsilon \}$. $A \rightarrow BC, B \rightarrow CA \mid b, C \rightarrow AB \mid a$	4
d	Draw the NFA accepting the language generated by the grammar with productions $S \rightarrow abA \mid bB \mid aba, A \rightarrow b \mid aB \mid bA, B \rightarrow aB \mid aA$	4
5 a	Define PDA. Construct PDA for $L = \{ ww^R \mid w \in \{a,b\}^* \}$. Show by IDs the string abaaba is accepted.	6
b	Let L be $L(M1)$ for some PDA with empty stack $M1 = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, \phi)$, prove that there exists an final state PDA M2 such that $L = L(M2)$. Construct empty stack PDA to accept $L = \{ w \mid w \in \{a,b\}^* \text{ and } w \text{ is palindrome} \}$. Convert it into equivalent final state PDA.	4
c	What are steps to be followed while finding an equivalent PDA by empty stack for the given CFG. Find the equivalent PDA to the CFG with productions $S \rightarrow aSb \mid bSa \mid abS \mid baS \mid Sab \mid Sba \mid \epsilon$. Show that the string abaabb is generated by the given CFG and it is accepted by its equivalent PDA.	6

	OR	
6 a	State and prove pumping lemma for context free languages. Show that $L = \{ww \mid w \in \{a,b\}^*\}$ is not context free.	6
b	Let $L_1 = \{a^i b^j c^k \mid i < j\}$ and $L_2 = \{a^i b^j c^k \mid i < k\}$. Show that L_1 and L_2 are context free but $L_3 = L_1 \cap L_2$ is not context free.	6
c	Decide whether the following languages are CFL or not. i. $L = \{a^n b^m a^m b^n \mid m, n \geq 0\}$. ii. $L = \{xayb \mid x, y \in \{a,b\}^* \text{ and } x = y \}$.	4
7 a	Design Turing Machine to accept the language $L = \{w \mid \varepsilon \{a, b\}^* \text{ and } N_a(w) = N_b(w)\}$. Trace the machine for the string <i>baabab</i> .	6
b	Let x and y are two positive integers represented using unary notation. Design a Turing Machine that computes the function $f(x,y)$, where $x, y \in 1^+$ $f(x, y) = x - y \quad \text{if } x > y$ $f(x, y) = y - x \quad \text{if } y > x$ $f(x, y) = 0 \quad \text{if } x = y$	6
c	If L_1 and L_2 are recursively enumerable languages over Σ , then show that $L_1 \cup L_2$ is also recursively enumerable languages.	4
	OR	
8 a	Design Turing Machine to perform the string reverse operation over $\Sigma = \{a, b, c\}$. Use Instantaneous Descriptions to show operation on $w = abac$.	8
b	Write a note on i. Multi tape Turing Machine ii. Non deterministic Turing Machine	6
c	Prove that every regular language is also a recursively enumerable language.	2
9 a	Construct the Linear Bounded Automata for the language $L = \{ww^R \mid w \in \{a, b\}^*\}$. Trace the machine for the string <i>aabbaa</i> .	8
b	Define Context Sensitive Grammar. Give Context Sensitive Grammar to generate the language $L = \{a^n b^n c^n d^n \mid n \geq 1\}$. Show that the string <i>aabbccdd</i> is generated by the grammar.	4
c	Write a note on Chomsky hierarchy.	4
	OR	
10 a	Define Unrestricted Grammar. Give Unrestricted Grammar to generate the language $L = \{w \mid \varepsilon \{a, b, c\}^* \text{ and } N_a(w) = N_b(w) = N_c(w)\}$. Show that the string <i>acabcb</i> is generated by the grammar.	8
b	Write a note on undecidability.	4
c	If the given language L is Context Sensitive Language, then L is recursively enumerable language. Prove this statement.	4

Q.P.Code

Course code: 21CS54

Course Title: Theory of Computation

PART-A

Q.No	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10
BT	1	2	3	1	2	3	2	1	1	2
COs	1	1	2	1	2	2	2	1	2	1
Q.No	1.11	1.12	1.13	1.14	1.15	1.16	1.17	1.18	1.19	1.20
BT	3	4	4	4	1	3	3	2	1	4
COs	2	3	3	2	1	3	2	1	2	3

PART-B

Question No		BT level	COs Addressed		Question No		BT level	COs Addressed
2	a	2	1		3	a	3	2
	b	3	2			b	2	2
	c	4	3			c	4	3
	d	4	3			d		
4	a	3	3		5	a	3	3
	b	4	3			b	2	2
	c	4	4			c	2	3
	d	3	4			d		
6	a	2	2		7	a	4	4
	b	3	3			b	4	4
	c	3	4			c	3	2
	d					d		
8	a	4	4		9	a	4	4
	b	3	2			b	3	2
	c	3	2			c	1	1
	d	4	1			d		
10	a	3	3			a		
	b	2	1			b		
	c	2	1			c		
	d					d		

Signature of Scrutinizer

Name:

Signature of Chairman

Name: